

Article

Genome-Resolved Metagenomics Suggests Site-Specific Microbial Adaptations in Urban Soils Co-Contaminated with Hydrocarbons and Heavy Metals

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Abstract

This study explores the physicochemical properties and microbiological community structure of oil-contaminated soils from Midrand and Roodepoort, South Africa. Due to sample pooling, the analysis provides a composite profile for investigating site-specific microbial adaptations rather than replicated ecological inference. The soils of Midrand exhibited acidity (pH around 5.5–5.9), elevated levels of heavy metals (e.g., Zn exceeding 1000 mg/kg), and the presence of 5–6 ring polycyclic aromatic hydrocarbons (PAHs). The soils in Roodepoort exhibited a near-neutral pH (about 6.2–7.2), characterized by specific metal concentrations (e.g., Cr exceeding 150 mg/kg) and an elevated presence of four-ring polycyclic aromatic hydrocarbons (PAHs). Metagenomic analysis indicated distinct microbial communities: *Pseudomonas* spp. were prevalent in Midrand, while *Bacillus* spp. were dominant in Roodepoort. Correlation analysis suggested connections between pollutants and microbial taxa; however, these findings are tentative. Recovered metagenome-assembled genomes (MAGs) indicated genetic potential for polycyclic aromatic hydrocarbon (PAH) degradation in Midrand and for metal resistance in Roodepoort. The findings suggest that localised pollution profiles are associated with unique microbial community structures and genetic potentials, providing a genomic basis for proposing site-specific bioremediation strategies. The research underscores the necessity for measures that take into account pollutant composition, soil pH, and microbial adaptation.

Keywords: soil pollution; polycyclic aromatic hydrocarbons; heavy metals; shotgun metagenomics; metagenome-assembled genomes (MAGs); bioinformatics



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1. Introduction

Soil is an essential environmental element, facilitating ecosystem functioning, nutrient cycling, and various biological populations [1,2]. Nonetheless, anthropogenic activities, such as industrialization and urbanization, result in soil contamination by pollutants like polycyclic aromatic hydrocarbons (PAHs) and heavy metals [3–5]. These pollutants can persist, lowering soil fertility, impeding microbial activity, and threatening human and ecological health [6]. Microorganisms are key agents in the natural remediation of

contaminants. Hydrocarbon degradation is frequently associated with bacterial phyla such as Proteobacteria, Firmicutes, and Actinobacteria [7,8]. Oil contamination often leads to a decrease in the overall diversity of soil microbes, as it promotes specialised degraders possessing essential functional genes for the degradation of hydrocarbons and PAHs [9,10]. A limited fraction of these microbial communities can be isolated and analysed using traditional culture-based techniques; however, recent advancements in shotgun metagenomic sequencing permit comprehensive characterization of both taxonomic and functional attributes [11–13]. Consequently, revealing genetic capabilities for hydrocarbon breakdown and other metabolic pathways significant to bioremediation.

Despite global advancements, a considerable knowledge gap remains in comprehending the functional adaptations of native microbial communities to simultaneous hydrocarbon and heavy metal contamination, a common real-world scenario. Although several studies combine physicochemical and metagenomic data [14–17], there exists a need for genome-resolved analyses that clearly link contaminant profiles to the functional genetic apparatus of microbial consortia to facilitate predictive bioremediation. This study fills a major knowledge gap in how microbial communities adapt functionally to simultaneous pollution from heavy metals and polycyclic aromatic hydrocarbons (PAHs), which is a scenario that occurs often in urban environments but has received very little research [18–20]. In contrast to the fact that earlier research has frequently investigated these contaminants in isolation [21–23], we present a comprehensive, genome-resolved metagenomic analysis that establishes a connection between certain contaminant profiles and particular microbial consortia and their genetic structures.

This study examines the adaptations of microbial communities in oil-contaminated urban soils from two locations in South Africa to elucidate the links between mixed pollutants and microbial populations. We integrate physicochemical characterization with comprehensive shotgun metagenomics and correlation analysis. The specific objectives are to: (i) characterize the physicochemical properties (pH, PAHs, heavy metals, etc.) of soils from Midrand and Roodepoort; (ii) profile the taxonomic composition and recover metagenome-assembled genomes (MAGs) from native bacterial communities; (iii) explore relationships between environmental factors and microbial community structure; and (iv) characterize the genetic potential for stress response, metal resistance, and hydrocarbon degradation to inform hypotheses about each site's bioremediation capability.

2. Materials and Methods

2.1. Site Description and Soil Sampling

The soil samples were collected from two suburban locations in Gauteng Province, South Africa, Roodepoort and Midrand, located 32.5 km from each other. As depicted in Figure 1, both these locations were characterized by contamination with used car engine oil and other hydrocarbons from the informal vehicle service workshop located next to the taxi ranks. The sampling points were evenly distributed (about 70 cm apart) to cover the entire area contaminated with oil. The soil samples were collected in triplicate using a sterile spatula from two levels, the surface layer (0–10 cm) and the subsurface layer (10–20 cm), then transferred into sterile zipper bags. All samples were transported in cooler boxes and stored at 4 °C at the University of South Africa (UNISA) Science laboratories in Roodepoort. For physicochemical and whole-genome shotgun metagenomics analysis, the soil samples were pooled according to the sampling location and depth. The resulting samples for Roodepoort: surface layer (0–10 cm), subsurface layer (10–20 cm), Midrand: surface layer (0–10 cm), and subsurface layer (10–20 cm) were given the following designations: Rood_10 cm, Rood_20 cm, Mid_10 cm, and Mid_20 cm, respectively. Although pooling may introduce limitations by obscuring site-specific variations, the decision to pool samples was

made to enhance sequencing depth, allowing for a greater number of high-quality reads per sample. Nonetheless, it is recognized that pooling diminishes the effective sample size ($n = 4$), thereby constraining the statistical power for inferential analysis and obscuring fine-scale spatial variability. The resultant dataset is thus more appropriate for exploratory, descriptive metagenomic profiling than for rigorous statistical inference [24–26].

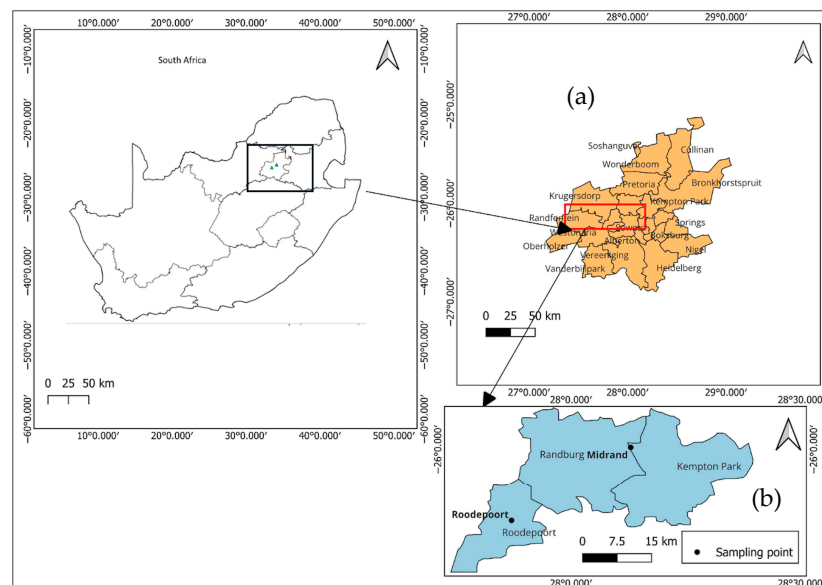


Figure 1. QGIS map showing the two sampling site locations in Gauteng, South Africa: (a) Roodepoort ($26^{\circ}9'32''$ S, $27^{\circ}52'15''$ E) and (b) Midrand ($25^{\circ}59'57''$ S, $28^{\circ}7'46''$ E). The map was created using QGIS software (version 3.44.0; URL: <http://www.qgis.org/>).

2.2. Physicochemical Analysis of Soil Samples

The pH of the soil samples was measured following the ISO 10390:2021 standard, which ensures reproducibility and comparability [27]. Briefly, 2 g of soil was placed in 10 mL of deionized water (dH_2O) and allowed to stand for 30 min before being measured using Mettler Toledo (Seven Direct SD20) (Mettler-Toledo International Inc., Greifensee, Switzerland). For subsequent analysis, the soil samples were cleaned of any plant residue and finely ground using a pre-sterilized coffee grinder (SCG-250), then passed through a 250 μm sieve.

2.3. CHNS Analysis of the Soil Samples

Samples were dried in a preheated oven at 55°C for 24 h in preparation for elemental analysis. The characterization of the carbon, hydrogen, nitrogen, and sulphur (CHNS) composition of the samples was performed at the University of Witwatersrand (South Africa) using the CHNS Vario EL Cube instrument (Elementar, Germany) with sulphanilamide as the standard. Approximately $[(1-1.5) \pm 0.01 \text{ mg}]$ of each sample was weighed using a Labotec precision balance (YP series), packed into a tin capsule, and then placed in the automated sampler of the device. The data were converted from weight to atomic percentage [28].

2.4. Metal Analysis

Twenty-five milligrams of samples were weighed utilizing an analytical balance (Mettler-Toledo XP205, Powai, Mumbai, India) and transferred to a microwave digestion vessel. Nine millilitres of Supra Pure (Merck Chemicals, Darmstadt, Germany) HNO_3 and 3 mL of Supra Pure (Merck, Darmstadt, Germany) HCl were added to each vessel. The vessels were then transferred to a microwave digester, and the temperature was raised to

180 °C for 10 min and held at the same temperature for an additional 10 min. Afterward, the samples were quantitatively transferred to 50 mL volumetric flasks and made up to the 50 mL mark using ultrapure water with a conductivity of 18.2 Mohr/cm. The samples were centrifuged at 6000 RPM for 10 min. This was followed by a 20× dilution by pipetting 500 µL of the sample into 10 mL of 2% HNO₃ (Merck, Darmstadt, Germany). To ensure quality, AMIS0373 (Amis, Modderfontein, South Africa) was prepared in the same way as the samples. The instrument was calibrated using 100 mg/L of NIST traceable stock (de Bruyn Spectroscopic Solutions, Kyalami, South Africa) as standards (0 mg/L, 0.1 mg/L, 1 mg/L, and 10 mg/L). The samples and calibration standards were analyzed using a NexION 300D Quadrupole-ICP-MS (PerkinElmer, Waltham, MA, USA) in collision mode. The linear regression line produced by the calibration standards (instrument response against concentration) was utilized to determine the element concentrations in the samples [29].

2.5. Polycyclic Aromatic Hydrocarbons (PAHs) in the Soil and Analysis

Approximately 10 g of the sample was weighed and mixed with a deuterated internal standard. The mixture underwent an accelerated solvent extraction process using activated alumina and Florisil for in-cell cleanup. Two processing steps were implemented: (1) dilution with toluene and (2) dispersive clean-up using magnesium sulphate, primary-secondary amine exchange material (PSA), and C18. The extraction process was carried out using a waste stream method. The extraction amount was reduced to 5 g, and the volume of activated alumina and Florisil was increased significantly. The resulting extracts were then evaporated and reconstituted into toluene before instrumental analysis. The instrumental analysis was conducted utilizing a Pegasus 4D gas chromatograph linked with a time-of-flight mass spectrometer (GC-TOFMS) (Leco, St. Joseph, MI, USA) and an Rxi[®]-XLB column (30 m, 0.25 mm ID, 0.25 micrometres df). The instrument parameters were derived from a Restek application note on PAHs and PCB congeners on Rxi[®]-XLB (Restek Corporation, Bellefonte, PA, USA). A non-matrix-matched 12-point calibration curve was established [30]. The levels of contamination with PAHs and the possible sources of contamination were also determined using appropriate formulae [31,32].

2.6. Whole-Genome Shotgun Metagenome Profiling

2.6.1. DNA Extraction and Sequencing

Genomic DNA was extracted from soil samples, according to the manufacturer's protocol, using the NucleoSpin soil extraction kit (Macherey-Nagel GmbH & CO. KG Dueren, Germany). The purity of the DNA was assessed at an absorbance ratio of 260 to 280 nm using the Bio Drop spectrophotometer (Biochrom, Cambridge, UK), and the concentration was measured using a QUBIT fluorometer (ThermoFisher, Waltham, MA, USA). The extracted DNA was stored in cryotubes at −20 °C until it was sent to Inqaba Biotechnical Industries (Pty) Ltd. (Pretoria, South Africa) for whole-genome shotgun metagenomics sequencing on a NextSeq 500 platform (Illumina, San Diego, CA, USA).

2.6.2. Bioinformatics

The raw reads from sequencing were quality checked (base quality score, adapter contamination, GC content, and sequence length distribution) using FastQC v0.1.1.9. All subsequent read processing and assembly steps were performed within the KBase narrative interface (<https://narrative.kbase.us>) (accessed on 8 December 2025). Filtering and trimming of reads were carried out using Trimmomatic v0.39 (parameters: ILLUMINACLIP: TruSeq3-PE. 2:30:10, LEADING:3, TRAILING:3, SLIDINGWINDOW:4:15 and MINLEN:50 [33–35]). Post-trimming FastQC analysis was used to confirm the quality assessment of the trimmed reads. Metagenomic assembly was carried out separately for every individual sample using MEGAHIT v1.2.9, following default settings [36,37].

MetaQUAST v5.0.2 was used to assess assembly statistics (total contig length, largest contig size, including N50 and GC content), and contigs longer than 500 bp were retained for downstream analysis [38,39]. The RAST toolkit in KBase was used for gene prediction and annotation of the assembled contigs [40]. This enabled identification of all protein-coding sequences (CDSs) and assigned putative functions based on homology to the SEED database [41]. DRAM (Distilled and Refined Annotation of Metabolism) v1.4 was used to analyze metabolic potential on both raw metagenome assemblies and MAGs, with the focus on KEGG modules for stress and hydrocarbon degradation [42]. Metagenome-assembled genomes (MAGs) were recovered using a single-binner approach with MaxBin2 v2.2.7 (default parameters). MaxBin2 was selected for its computational efficiency and proven reliability in producing high-quality, conservative bins suitable for our downstream comparative metabolic analysis. The resulting bins were quality checked with CheckM v1.1.3 (based on the existence of single-copy marker genes, providing estimates of contamination and completeness) [43,44]. Following the MIMAG standard, only medium- and high-quality MAGs with more than 50% completeness and contamination less than 10% were kept [43,45]. GOTTCHA2 v2.1.6 was used for taxonomic profiling against RefSeq complete genomes [46,47]. Alpha (α) and beta (β)-diversity (Bray–Curtis + PCoA) were determined using the vegan package in R (v4.1.0) [48]. Kruskal–Wallis and Wilcoxon rank-sum tests were employed for statistical and differential abundance ($p < 0.05$). Taxonomic profiles, MAG characteristics, and functional annotations were used to conduct a comparative analysis between samples [49]. Genes involved in stress response and hydrocarbon degradation were assessed for differential abundance across the samples.

2.6.3. Statistical Analysis

GOTTCHA2 v2.0 was used for taxonomic profiling, and species-level assignments were kept for further examination. TSV output files for four samples, Rood_10 cm, Rood_20 cm, Mid_10 cm, and Mid_20 cm, were examined to extract relative abundance information. The mean relative abundance across all samples was used to determine the top 20 most prevalent bacterial species. Environmental parameters comprised 10 heavy metals (Cr, Mn, Co, Ni, Cu, Zn, As, Cd, Hg, and Pb), 3 elemental composition parameters (carbon, hydrogen, and nitrogen), and 16 polycyclic aromatic hydrocarbons (PAHs: Nnb, AY, AE, Fnb, Pnb, Anb, FLnb, Pynb, B[ghi]f, BaA, Cnb, BaF, BaP, IP, ghi, and DA). Quadruplicate samples were used for each measurement. All statistical analyses were conducted using R version 4.3.1. Given the pooled sampling design's effective sample size of $n = 4$, inferential statistical power is inherently limited. Consequently, all ensuing correlation studies must be regarded with caution as exploratory and hypothesis-generating rather than as definitive evaluations of statistical significance. The limited sample size heightens the likelihood of both Type I and Type II errors, and correlation coefficients, even with $|\rho| > 0.8$, may occur by mere coincidence. Consequently, these studies present their findings as observed co-occurrence patterns or trends, necessitating future validation with larger, non-aggregated datasets. With this significant stipulation, Spearman's rank correlation coefficient (ρ) was employed to assess potential associations between microbial taxa and environmental factors, owing to the non-normal distribution of microbial abundance data. Correlations with $|\rho| > 0.8$ were noted as strong associations for descriptive purposes, while statistical significance was evaluated at $\alpha = 0.10$. Correlation tests were conducted with `cor.test` with `exact = FALSE` to address ties in the dataset.

Correlation heatmaps were produced using the `corrplot` software (Version 0.95), with colour gradients indicating correlation strength (blue: $\rho = -1$, white: $\rho = 0$, and red: $\rho = +1$). Scatter plots illustrated the association between taxonomic abundance and environmental responsiveness, with the point size reflecting the mean relative abundance and the colour

intensity denoting the mean absolute correlation strength. Augmented taxonomic labels incorporated the two primary environmental correlates for each taxon, utilising the ggrepel package to mitigate label overlap. Known PAH-degrading bacterial taxa were identified through literature curation, including genera such as *Pseudomonas*, *Mycobacterium*, *Rhodococcus*, *Sphingomonas*, *Bacillus*, and related species. Specific associations between these taxa and PAH compounds were analyzed separately to identify potential bioremediation candidates.

3. Results and Discussion

3.1. Physico-Chemical Characterization and Its Impact on Soil Ecosystems

3.1.1. Sampling and pH Profiles of Oil-Contaminated Soils

The soil samples obtained from oil-contaminated locations in Midrand and Roodepoort, adjacent to informal auto workshops in Gauteng Province, South Africa, displayed unique physicochemical characteristics. The pH study revealed that the Midrand samples were unequivocally acidic, whereas the Roodepoort samples were neutral to marginally basic. A discernible pattern was noted, indicating that pH levels were reduced at 0–10 cm depths in both locations, whereas the reverse was observed at 10–20 cm depths.

3.1.2. Polycyclic Aromatic Hydrocarbon Analysis and Quantification

The analysis showed that there was a significant difference in the levels of PAH contamination between the two locations (Figure 2). The soils of the Midrand region were distinguished by much higher overall concentrations of polycyclic aromatic hydrocarbons (PAHs), with high-molecular-weight (HMW) 5- and 6-ring compounds predominating. Based on this profile, it appears that petroleum pollution is persistent and severe. The soils of Roodepoort, on the other hand, had a lower overall PAH load and a higher proportion of compounds with four and five rings, which aligned more closely with combustion-related inputs. All of the soils that were sampled were in the category of being seriously polluted with PAHs, as determined by the recognised classifications for the severity of pollution.

The application of the feature ratios approach, namely $FLA/(FLA + PYR)$ and $BaA/(BaA + CHR)$, can yield preliminary insights into PAH sources, indicating that the contaminants may originate from both petroleum and combustion sources, in alignment with the historical contamination from used engine oil at the locations. A ratio of less than 0.4 indicates a potential petroleum source, while a ratio between 0.4 and 0.5 suggests a possible source of fossil fuel combustion, and a ratio greater than 0.5 points to a coal/wood/grass combustion source. A ratio of less than 0.2 for $BaA/(BaA + CHR)$ means that the source is petroleum, a ratio between 0.2 and 0.35 means that the source is mixed, and a ratio of more than 0.35 means that the source is combustion [50,51]. The $FLA/(FLA + PYR)$ ratios for Mid_10 cm, Mid_20 cm, Rood_10 cm, and Roodepoort were 0.24427481, 0.26153846, 0.29113924, and 0.33333333, respectively. This means that the source was oil. For $BaA/(BaA + CHR)$, the ratios for Mid_10 cm, Mid_20 cm, Rood_10 cm, and Rood_20 cm were 0.52727273, 0.53191489, 0.54347826, and 0.49019608, respectively, indicating a combustion source. Utilizing the feature ratios method of $FLA/(FLA + PYR)$ and $BaA/(BaA + CHR)$ revealed that the PAHs may stem from both petroleum and combustion sources, as the sampling locations have been characterized by years of pollution from used car engine oil. The predominance of 5- and 6-ring PAHs in Midrand suggests chronic petroleum pollution, whereas Roodepoort's 4-ring PAHs align with combustion-related inputs. Nonetheless, the interpretation of these diagnostic ratios in aged, biologically active soils necessitates caution. These ratios are predicated on the comparative stability of several PAH compounds under optimal conditions. In polluted environments with active microbial communities, as demonstrated in this work, differential biodegradation can substantially modify these ratios over time. Consequently, the calculated ratios may indicate a mixture

of the initial pollution source signature and later in situ microbial modification, rather than solely the source itself. This limitation emphasizes that although ratio analysis is a valuable technique, it must be evaluated in conjunction with other data [52–54].

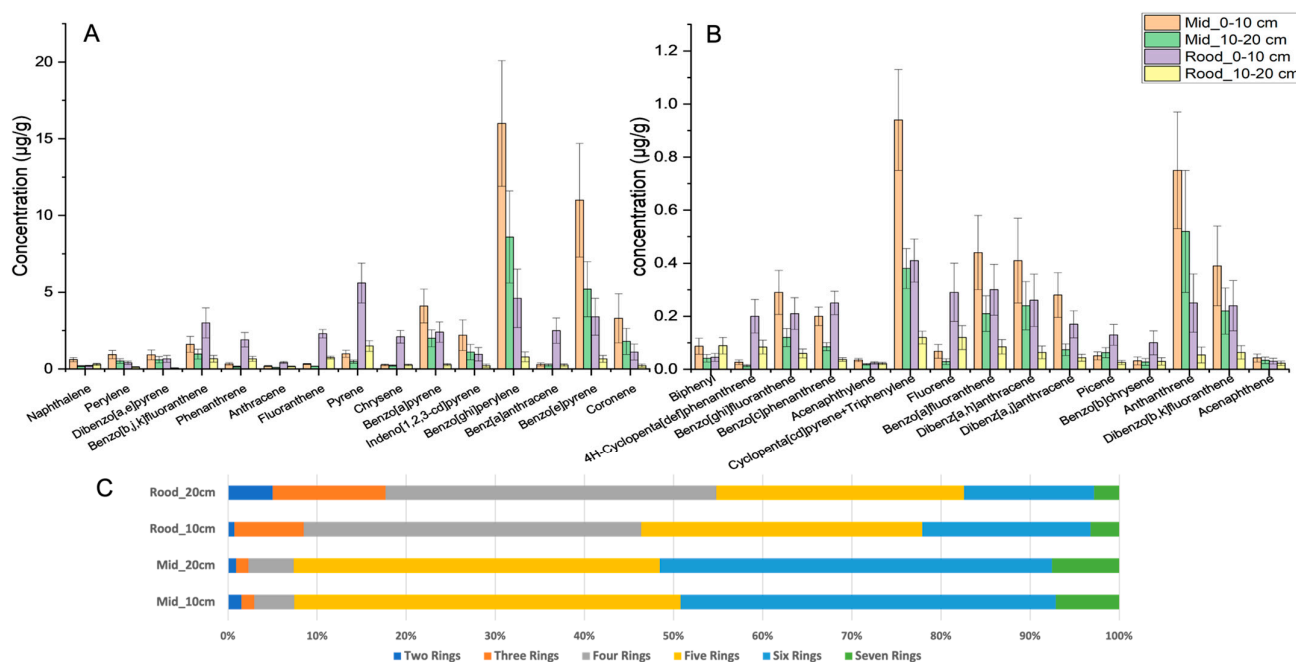


Figure 2. Concentrations of 30 polycyclic hydrocarbons (PAHs) in the soil samples: (A) PAHs with concentrations between 0 and 20 µg/g. (B) PAHs with concentrations between 0 and 1.2 µg/g. (C) The proportion of specific PAH compounds relative to the overall number of PAHs present in soils.

3.1.3. Elemental Composition (C, H, N, S) in Oil-Contaminated Soils

Elemental analysis demonstrated significant disparities in organic matter concentration between the sites (Figure 3). The surface soil at Midrand displayed elevated quantities of carbon and hydrogen, indicative of a significant influx of petroleum-derived hydrocarbons. Nitrogen was identified solely in this sample, although sulphur remained beneath detection thresholds in other samples. The markedly reduced carbon and hydrogen levels in Roodepoort soils suggest a diminished direct hydrocarbon load. The pervasive deficiency of nitrogen and sulphur in the majority of samples indicates restricted nutrient cycling and possible constraints on microbial activity and plant growth in these degraded soils [55,56].

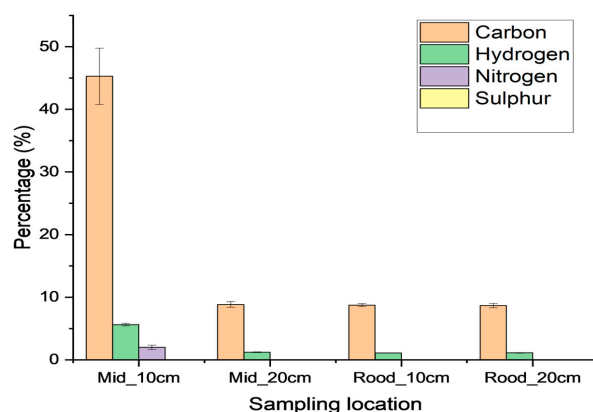


Figure 3. Elemental composition of oil-contaminated soils from Midrand and Roodepoort, showing concentrations of major elements (carbon, hydrogen, nitrogen, and sulphur). Values are expressed in percentages, highlighting variations across sampling depths (0–10 cm and 10–20 cm).

3.1.4. Heavy Metal Analysis and Comparison to International Standards

Heavy metal analysis indicated significant and localised contamination (Figure 4). The soils of Midrand were notably enriched with zinc and manganese, whilst the soils of Roodepoort exhibited heightened concentrations of chromium, copper, and lead. A prevalent vertical tendency was noted for the majority of metals, exhibiting elevated concentrations in the surface layer (0–10 cm), presumably attributable to recent deposition and diminished mobility. Exceptions, such as manganese and cobalt, which showed a rise with depth, may suggest distinct leaching behaviours or historical deposition patterns. Mobility and leaching potential, weathering, solubility, and microbial interactions may be the reasons for lower concentrations at the deeper depths [57,58]. Comparisons with international soil quality guidelines (WHO, Sweden, United Kingdom) indicated several significant exceedances, affirming a critical ecological danger (Table 1). Zinc and chromium concentrations in Midrand and Roodepoort, respectively, were exceedingly elevated, far exceeding the most rigorous standards. Concentrations of lead, copper, and nickel surpassed health-based criteria at both sites, though to differing extents. Cadmium levels were predominantly within acceptable limits, except for one surface sample from Midrand. The similarities clearly categorise the sampled locations as severely contaminated and environmentally degraded. These comparisons suggest that the sampling sites were heavily contaminated and in a deteriorating state [59–61].

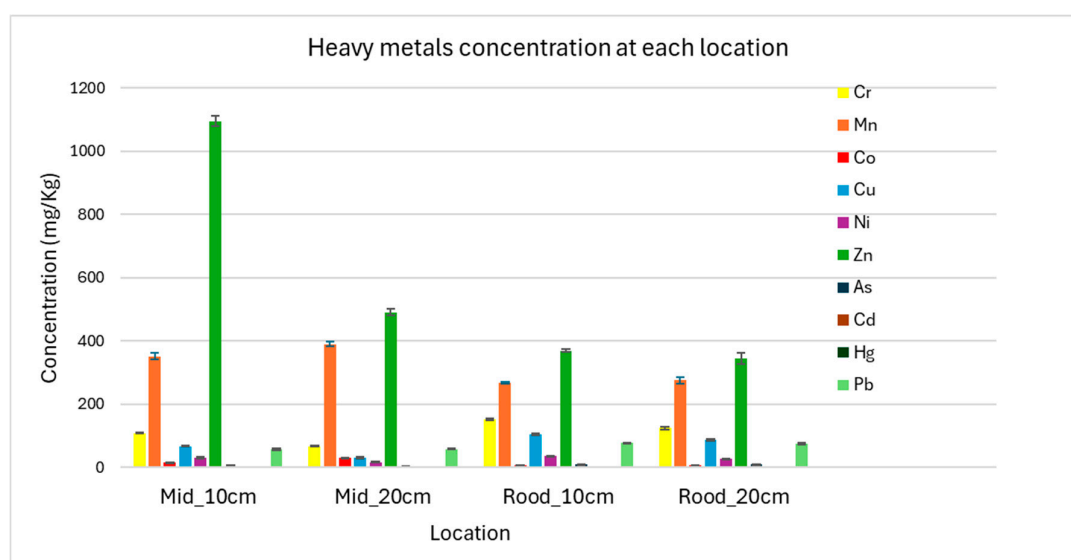


Figure 4. Heavy metal concentration of oil-contaminated soils from Midrand and Roodepoort, showing concentrations of major heavy metals (Zn, Mn, Cr, Co, Pb, Cu, Ni, etc.) detected via ICP-MS. Values are expressed as mg/kg of dry soil mass, highlighting variations across sampling depths (0–10 cm and 10–20 cm).

Table 1. Soil quality criteria for heavy metal concentrations in soil (mg/kg). The limits are based on three standards: WHO, Sweden, and the United Kingdom.

	Mid-10 cm	Mid-20 cm	Rood-10 cm	Rood-20 cm	** WHO	* Sweden	* United Kingdom
Cadmium (Cd)	0.51	0.29	0.33	0.28	0.31	0.4	3
Chromium (Cr)	108.33	67.31	151.67	124.67	8	60	400
Copper (Cu)	67.43	30.74	104.77	86.31	0.5	40	135
Nickel (Ni)	30.39	16.39	34.9	27.61	20	30	75
Lead (Pb)	56.22	57.95	77.57	74.86	13.0	40	300
Zinc	1095	491	369	345	1.5	—	—

* Source: ECDGE [62]. ** Source: (World Health Organization (WHO), [63]).

3.1.5. Integrated Environmental Interpretation

The implications of the above-mentioned physicochemical properties, particularly high levels of PAH pollution, high concentrations of heavy metals, PAHs of mixed origins, as well as pH shifts, play a critical role in shaping soil ecosystems [64–66]. Acidic pH, elevated concentrations of heavy metals (such as Zn, Pb, Cu, Ni), dominance of 5- and 6-ring PAHs, combined with the total PAH concentration in Midrand, create a very toxic environment. The shift to acidic conditions ($\text{pH} < 6$), as supported by Jabbarov et al. [67], increases the solubility and bioavailability of heavy metals, which in turn can inhibit microbial activity and plant growth [68,69]. The low pH of the soil often reduces the availability of essential nutrients to plants and microorganisms, limiting their growth and affecting hydrocarbon degradation [70–72]. Some bacteria, such as *Pseudomonas* and *Bacillus*, are less active in acidic soils, which may slow down oil biodegradation [70–72]. The high concentrations of carbon and hydrogen, i.e., 45.28 and 5.633% for Mid_10 cm, suggest the presence of significant organic matter in this area. Organic matter serves as a carbon and energy source for microbial communities, supporting their growth and activity. However, the organic matter of hydrocarbon origin affects soil health, microbial health, and the overall ecosystem [73]. This type of environment only supports the growth of a few specialized types of plants and microbial populations [74–76]. For example, hydrocarbon-degrading bacteria (e.g., *Pseudomonas*, *Bacillus*) thrive in carbon-rich environments, enhancing bioremediation of oil-contaminated soils [75,76].

At high concentrations, heavy metals have been shown to inhibit plant growth by damaging root systems, compromising nutrient uptake, and leading to oxidative stress [77]. For instance, high Zn and Cu concentrations can cause the yellowing of leaves and underdeveloped plants [78]. The high concentrations of Zn and Mn can lead to the degradation of soil structure by causing soil compaction, rendering plant growth less suitable [77,78]. Moreover, the effects of elevated heavy metal concentrations lead to a loss of soil fertility and biodiversity by reducing the solubility of essential nutrients and inhibiting the growth of microorganisms and plants [79,80]. Severe petroleum pollution, as well as the dominance of 5- and 6-ring PAHs in Midrand, makes the soil highly toxic due to their extremely persistent complex structure and low solubility. They are known to pose greater risks to soil health, thus affecting microbial diversity and plant growth [81,82]. Benzo[a]pyrene (5-ring) and indeno[1,2,3-cd]pyrene (6-ring) are particularly more concerning due to their high toxicity and resistance to degradation [82].

In contrast, Roodepoort's neutral to slightly alkaline pH, lower $\Sigma 30$ PAHs of 34.450 $\mu\text{g/g}$, and the prevalence of moderately toxic and persistent 4-ring and, to a lesser degree, 5-ring PAHs from combustion sources may facilitate greater microbial diversity and biodegradability. Neutral to slightly alkaline conditions at Roodepoort are usually optimal for many hydrocarbon-degrading bacteria, including *Pseudomonas* and *Bacillus* [83]. PAHs such as pyrene (4-ring) and benzo[a]pyrene (5-ring) are known to be mutagenic and carcinogenic [84]. However, these PAHs are more readily degraded by microorganisms compared to higher-ringed PAHs [84,85]. Consequently, microbial communities in Roodepoort may be more active and diverse, with a higher proportion of PAH-degrading bacteria. For instance, *Pseudomonas* species are known to degrade 4-ring PAHs like pyrene [86,87]. Since these PAHs are more amenable to biodegradation, bioremediation efforts may be more effective in Roodepoort compared to Midrand, as microbial communities can degrade these compounds into less toxic intermediates. However, elevated concentrations of Cr, Cu, Pb, Zn, and Ni still pose ecological risks. The low concentrations of carbon and hydrogen in Roodepoort indicate poor organic matter content, which can negatively affect soil structure, nutrient cycling, and microbial activity. The lack of nitrogen and sulphur in most samples suggests a constrained microbial activity and plant growth [55,56]. Overall,

based on these findings, both locations were severely contaminated, although at varying concentrations, emphasising an urgent need for site-specific bioremediation by either pH adjustments in Midrand and/or biostimulation in Roodepoort to enhance the activity of hydrocarbon-degrading microbes to mitigate ecological and health risks. Neutral to alkaline conditions reduce heavy metal solubility, lowering their toxicity and creating a more favourable environment for microbial diversity and activity, as well as plant survival [80].

3.2. Metagenome Assembly and MAG Recovery

Using MEGAHIT v1.2.9, each de novo assembly of the four samples of oil-contaminated soil metagenome produced high-quality assemblies (Supplementary Table S1; MetaQUAST summary statistics of MEGAHIT assemblies (contigs ≥ 1 kb)). With 3036–4988 contigs larger than 1 kb retained per sample, the overall assembled length varied from 9.85 Mb to 26.41 Mb. Mid_10 cm (7889 bp) and Mid_20 cm (8980 bp) had the greatest contig N50 values, indicating better assembly continuity in surface and subsurface Midrand samples, respectively. Of all the samples, Mid_10 cm assembly had the longest contig (350827 bp) and the only contigs larger than 100 kb ($n = 15$). The GC content differed significantly between the sites (Roodepoort: 42.0–51.8% and Midrand: 56.9–63.5%), which is consistent with the predominance of low-GC Firmicutes in Roodepoort and high-GC Proteobacteria/Actinobacteria in Midrand. Following the MIMAG standards, MaxBin2 extracted 38 medium- to high-quality metagenome-assembled genomes (MAGs) from these assemblies. For example, CheckM analysis confirmed low contamination (0–8.4%) and high completeness (52.3–98.7%) in all retrieved bins (Supplementary Tables S2–S6; CheckM output and Figure S1; CheckM plot for Mid_20 cm).

3.3. Microbial Community Structure and Profiling

Significant differences in the microbial community structure between the two oil-contaminated sites were found using shotgun metagenomic sequencing and GOTTCHA2-based taxonomic profiling of quality-filtered reads. This revealed a distinct microbial community structure potentially shaped by site-specific physico-chemical conditions. The indices, Evenness, Shannon, Simpson, and Species richness, were measured following the Kruskal–Wallis statistical method. Alpha diversity indices (Shannon, Evenness, and Simpson) were the highest in Midrand (Mid_10 cm > Mid_20 cm), indicative of greater diversity, while the species richness index was relatively high at Roodepoort (Rood_10 cm < Rood_20 cm), showing higher species richness (Figure 5). The distinct microbial profiles between the sampling locations were confirmed using Beta diversity analysis via principal coordinate analysis (PCoA) in Supplementary Figure S2. The Mid_10 cm and Rood_20 cm samples had the highest Beta diversity compared to the Mid_20 cm and Rood_10 cm samples.

At the phylum-level taxonomic classification (Figure 6a), the three most dominant phyla were Proteobacteria (60% in Mid_10 cm, 55% in Mid_20 cm, 20% in Rood_10 cm, and 16% in Rood_20 cm), Firmicutes (5% in Mid_10 cm, 15% in Mid_20 cm, 70% in Rood_10 cm, and 54% in Rood_20 cm), and Actinobacteria (13% in Mid_10 cm, 20% in Mid_20 cm, 5% in Rood_10 cm, and 10% in Rood_20 cm). At the species level (Figure 6b), both sampling levels at Midrand were dominated by up to 40% *Pseudomonas* species, which includes *Pseudomonas aeruginosa*, *Pseudomonas citronellios*, and *Pseudomonas* virus and phages. This dominance was followed by *Mycobacterium* and *Mycolibacterium* species: *Mycolibacterium doricum*, *Mycobacterium intracellulare*, and *Mycobacterium chimaera*. Mid_10 cm had significant levels of *Enterobacter* species (*Enterobacter hormaechei* and *Enterobacter cloacae*), *Citrobacter rodentium*, *Cupriavidus gilardii*, etc. Rood_10 cm and Rood_20 cm were dominated by up to 40% and 60%, respectively, of *Bacillus cereus*, *Bacillus mobilis*, and *Bacillus mycoides*. Whereas

15–20% for both levels were occupied by *Bacillus* species (*Bacillus* sp. ABp14, *Bacillus* sp. FDAARGOS-235, *Bacillus thuringiensis*, *Bacillus toyonensis*, *Bacillus velezensis*) and *Bacillus* viruses (*Bacillus* virus 250, *Bacillus* AvesoBmore, *Bacillus* virus B4, etc.). Besides *Bacillus* species, about 7–15% of both sampling levels were *Pseudomonas* species (*Pseudomonas aeruginosa*, *Pseudomonas citronellois*, etc.) and *Pseudomonas* phages (*Pseudomonas* phage F-MX1987 sp/MX560 and *Pseudomonas* phage Phi1).

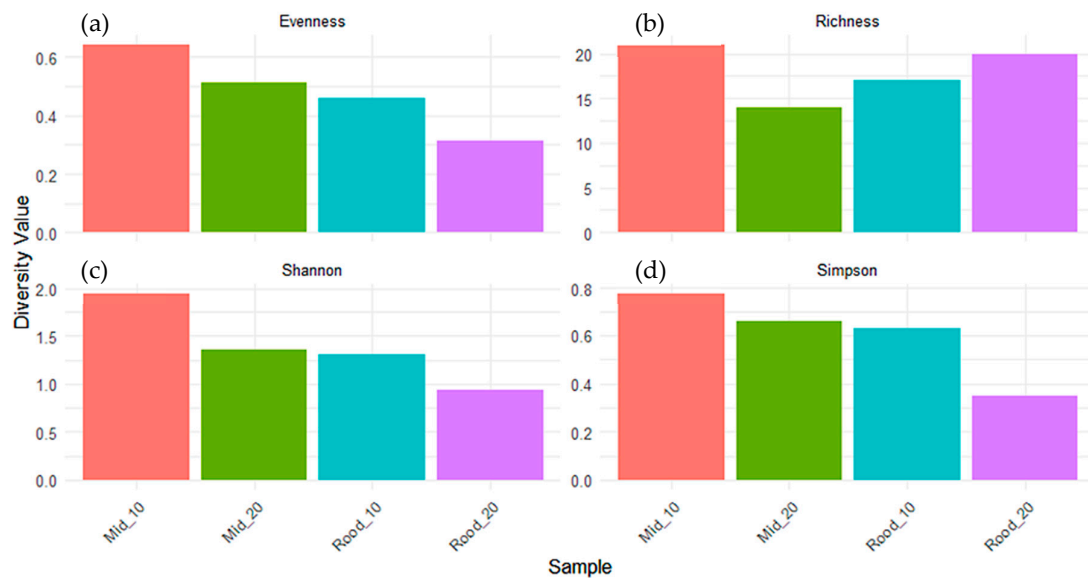


Figure 5. Alpha diversity indices of bacterial communities in oil-contaminated soils from Midrand and Roodepoort: (a) Evenness, indicating the total number of unique bacterial taxa detected. (b) Richness index, measuring how evenly taxa are distributed within the community. (c) Shannon index, evaluating species richness and evenness. (d) Simpson index, assessing dominance and diversity among microbial communities.

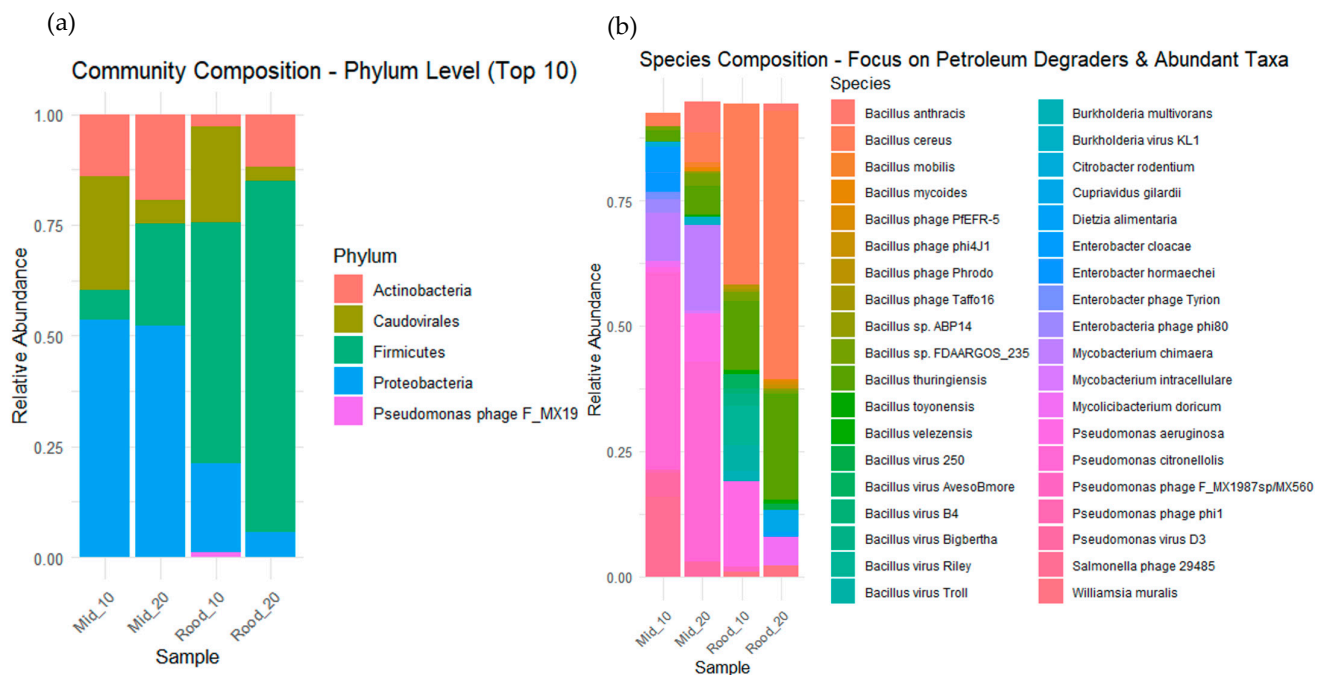


Figure 6. (a) Relative abundance of bacterial communities at the phylum level. (b) Relative abundance at the species level in oil-contaminated soil samples from Midrand and Roodepoort.

Figure 7 shows the taxonomic profiling at the strain level. *Pseudomonas* bacteria, including *Pseudomonas citronellolis* SJTE-3, *Pseudomonas citronellolis* P385, *Pseudomonas aeruginosa* C-NN2, and *Williamsia muralis* NBRC 105880, remarkably dominated the microbial population in Midrand. Both levels also had a significant relative abundance of *Mycobacterium chimaera* strain DSM 44623, *Mycobacterium chimaera* strain DSM FLAC0070, and *Mycobacterium doricum* strain DSM 44339 compared to the soil samples in Roodepoort. Roodepoort soils, on the other hand, showed far higher strain-level evenness and variety, with no strain above 20% relative abundance. Several strains of the *Bacillus cereus* group, including *B. thuringiensis* serovar konkukian str. 97-27, *B. cereus* AH187, ATCC 10987, ATCC 4342, NC7401, and several others (collectively, 40–45%), dominated the community. In addition, metal-tolerant strains of *Cupriavidus gilardii* CR3, *Enterobacter hormaechei* subsp. Xiangfan Gensis LMG27195, and *Enterobacter cloacae* were also significant, along with occasional contributions from *Mycobacterium* and *Mycolicibacterium* species (*M. chimaera* DSM 44623 and DSM FLAC0070, *Mycolicibacterium doricum* DSM 44339). The strain-level data show almost complete taxonomic differences between the two locations, with *Pseudomonas citronellolis-aeruginosa* lineages forming a single culture in acidic, high-PAH, metal-rich Midrand soils, while biosurfactant-producing *Bacillus cereus/thuringiensis* consortia dominate under the near-neutral, 4-ring-PAH and chromium-rich conditions in Roodepoort.

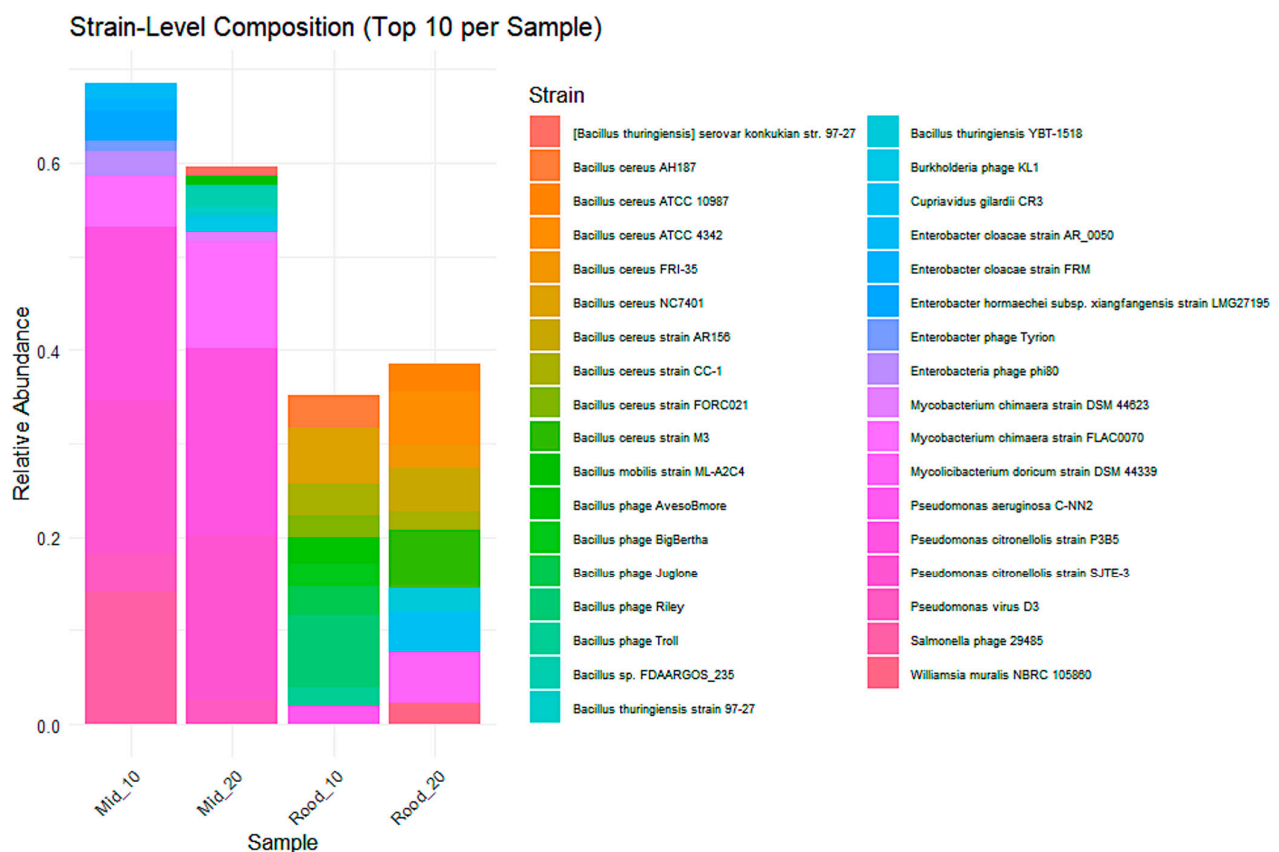


Figure 7. Strain-level distribution and relative abundance of bacterial communities in oil-contaminated soil samples from Midrand and Roodepoort.

The presence of these phyla, genera, and species, as reflected by alpha and beta diversity indices, suggests their ecological roles in surviving under extreme PAH contamination and their associated physico-chemical conditions as outlined in Section 3.1. The greater diversity in Midrand (Figure 4) was associated with an acidic pH (5.49), extreme levels of heavy metals, high carbon and hydrogen content, as well as severe PAH contamination,

which may have encouraged a diverse Proteobacteria-dominated community adapted to chronic oil contamination. This dominance is in accordance with other studies, where Proteobacteria were found dominant in oil-contaminated soils [88,89]. The prevalence of *Pseudomonas* in the acidic Midrand soils seemingly contradicts the common understanding that its growth is ideal at neutral pH. However, increasing data suggest that members of the genus *Pseudomonas* can demonstrate significant acid tolerance or adaptability in particular environmental and functional circumstances. *Pseudomonas citronellolis* YN-21 has been documented to sustain optimal heterotrophic nitrification activity at pH 5 while concurrently withstanding heightened metal ion stress, underscoring its physiological adaptability in acidic and metal-laden environments [22]. A *Pseudomonas* strain (MTS-1) that degrades polycyclic aromatic hydrocarbons (PAHs) exhibited robust growth and degradation of high-molecular-weight PAHs within a moderately acidic to neutral pH range (pH 5–8), though its activity was suppressed below pH 5, indicating a threshold for functional efficacy under heightened acidity [90]. In addition to laboratory investigations, environmental assessments of acid rock drainage (ARD) and streams affected by mining have consistently revealed the presence of acid-tolerant Proteobacteria, such as *Pseudomonas*, flourishing in low-pH, metal-contaminated environments, thereby reinforcing the ecological significance of acid adaptation within this genus [91,92]. These findings collectively contest the traditional perception of *Pseudomonas* as mostly neutrophilic and highlight its functional diversity and adaptability in acidic conditions.

The key genera, *Pseudomonas* and *Citrobacter*, have been shown to produce biosurfactants that effectively aid in the removal of hydrocarbons [93,94]. *Pseudomonas* species have been shown to possess several genetic and biochemical mechanisms to tolerate and resist high concentrations of heavy metals, including Zn and Mn [71]. The presence of metal efflux pumps, metal-binding proteins, extracellular polymeric substances (EPS) production, and extracellular sequestration adds to their ability to tolerate and proliferate under high metal concentrations, as reported in these studies [71,95]. *Pseudomonas aeruginosa* has been associated with the degradation of long-chain alkanes [96], *Mycobacterium* sp. with the degradation of monoaromatic compounds [97], *Bacillus licheniformis* with the degradation of polyaromatic compounds [98], and *Acinetobacter* sp. with the metabolization of aliphatic hydrocarbons [99], among others.

The higher species richness in Roodepoort is reflected in its carbon-rich, neutral pH, less severe 4-ring dominated, potential combustion-derived PAHs, which may encourage a Firmicutes-dominant community (e.g., *Bacillus* sp., *Bacillus cereus*, etc.). Between the sites, beta diversity (PCoA) uncovered significant community dissimilarities, likely as a result of pH differences, spikes in heavy metal levels (Midrand: Zn, Pb, while Roodepoort: Cu, Cr), and PAH sources (petroleum vs. combustion), amongst other factors. The dominance of Firmicutes over Proteobacteria is not a common occurrence in these environments, although it has been reported as one of the dominant phyla in hydrocarbon-contaminated soils [100,101]. However, it is worth noting that the dominance of these phyla can be influenced by various factors, such as the type and concentration of hydrocarbons present, as well as other environmental conditions [102,103].

Firmicutes were mostly constituted by the *Bacillus* genus, an observation consistent with other studies that have isolated and characterized *Bacillus* genera from oil-contaminated soils and explored their potential application in bioremediation [104–106]. The ability of *Bacillus* species to produce endospores for survival in harsh environments, as well as surface-active substances that increase hydrocarbon availability for efficient biodegradation, is one of the reasons for its dominance [107]. Furthermore, these species can tolerate and resist high concentrations of heavy metals, including zinc [108,109]. This is due to their genetic mechanisms, such as metal efflux pumps, metal-binding proteins, and

extracellular sequestration, that reduce the toxicity of heavy metals [110]. Additionally, their ability to form endospores and produce biosurfactants may promote the bioavailability of heavy metals and PAHs for bioremediation [111]. Therefore, the environmental conditions at Roodepoort may be uniquely enriched for this phylum over Proteobacteria, making Firmicutes better suited for any ecological function and potential Roodepoort-specific bioremediation efforts.

3.4. Observed Patterns Between Soil Properties and Microbial Communities: Ecological Insights

Observed co-occurrence patterns between the most prevalent bacterial strains and important physicochemical parameters in the oil-contaminated soils, such as heavy metals (As = arsenic, Ni = nickel, Pb = lead), macronutrients (carbon, nitrogen), and polycyclic aromatic hydrocarbons (PAHs: AE = acenaphthene, IP = indeno[1,2,3-cd]pyrene, GHI = benzo[ghi]perylene, Anb = anthracene, BaA = benz[a]anthracene, Cnb = chrysene), were examined following Kendall rank correlation analysis, which was displayed as a clustered heatmap (Figure 8). Taxa and parameters were grouped by similarity in connection patterns using hierarchical clustering, with correlation values ranging from -1 (strong negative, red) to $+1$ (strong positive, blue).

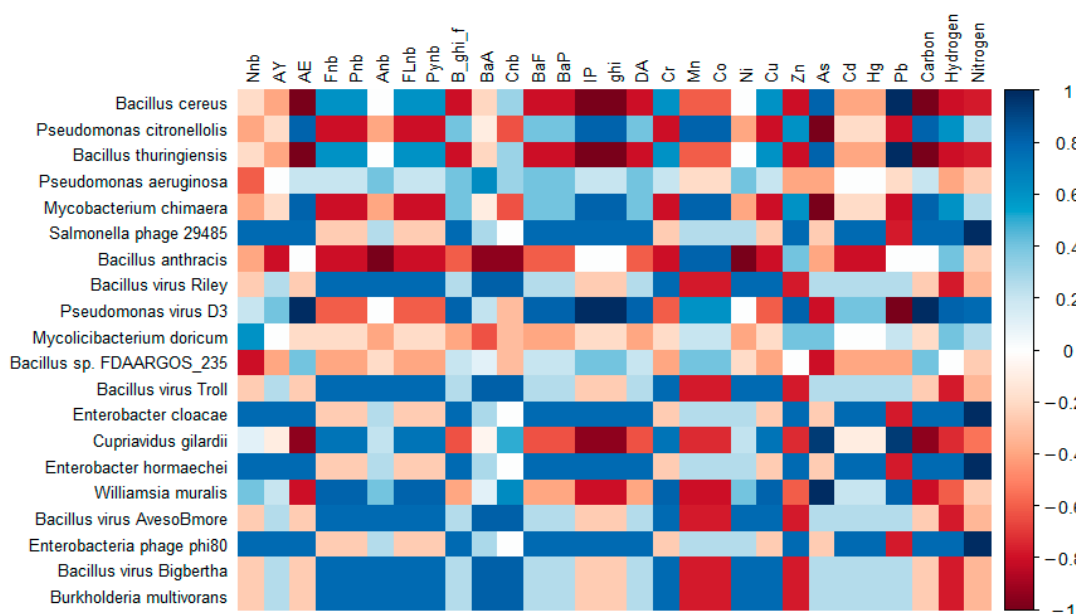


Figure 8. Kendall rank correlations analysis showing observed patterns between the top 20 dominant microbial taxa and physicochemical parameters in oil-contaminated soils.

Notable negative trends, which may suggest possible inhibition or avoidance under high contamination loading, were especially noticeable for several dominating taxa with particular PAHs and carbon content. *Bacillus cereus* and *Bacillus thuringiensis*, for example, showed notable negative associations with AE, IP, GHI, and total carbon ($\tau = -0.4$ to -0.6), indicating a pattern where Firmicutes strains appear less abundant in carbon-rich, HMW-PAH-dominated habitats such as the deeper layers of Midrand. In a similar vein, *Pseudomonas citronellolis* and As showed a high negative correlation, *Mycobacterium chimaera* and As showed a negative trend, and *Bacillus anthracis*, a close relative of *B. cereus*, showed a notable negative association with Anb, BaA, Cnb, and Ni. *Cupriavidus gilardii* showed a notable negative trend with AE, IP, GHI, and Pb, highlighting potential sensitivity to metals and PAHs. These trends align with the taxonomic alterations that have been observed: the acidic, PAH-rich environment of Midrand may restrict *Bacillus*-like organisms and benefit more tolerant *pseudomonads*.

On the other hand, taxa that appeared to thrive under particular contaminants were identified by notable positive associations; these taxa frequently exhibit adaptive features for bioremediation. Pb was positively associated with *Bacillus cereus* and *Bacillus thuringiensis*, which is consistent with the *B. cereus* group's known resistance to lead through efflux pumps and biosorption processes. *Salmonella* phage 29455, *Pseudomonas* virus D3, and *Enterobacteria* phage phi80 are examples of phage-related organisms that demonstrated positive correlations with nitrogen, which may suggest that viruses facilitate nutrient cycling in nitrogen-limited soils. Additionally, *Pseudomonas* virus D3 showed positive correlations with AE, IP, GHI, and carbon, suggesting a possible phage-mediated acceleration of *Pseudomonas*-driven PAH breakdown. *Enterobacter cloacae* and *Enterobacter hormaechei*, which are nitrogen-fixing or nitrogen-ammonifying enteric bacteria found in Roodepoort's neutral-pH, metal-enriched soils, showed a notable positive association with nitrogen. Notably, *Williamsia muralis* demonstrated a positive association with As, highlighting actinobacterial adaptations to metalloid stress, and *Cupriavidus gilardii* showed a positive association with As and Pb, consistent with its well-documented heavy-metal resistance (such as arsenic oxidation and lead sequestration).

These observed associations provide insights into the previously noted site-specific microbial adaptations. Negative associations between important degraders and HMW-PAHs and As in Midrand point to toxicity thresholds that may favour hardy strains of *Pseudomonas* and *Mycobacterium*, whose MAGs encode full PAH catabolic pathways even in the presence of inhibitory circumstances. As suggested by the abundant biosurfactant genes in their MAGs, *Bacillus* and *Cupriavidus* species in Roodepoort exhibit positive metal associations that highlight a community optimised for metal tolerance and biosurfactant-aided breakdown of lower-ring PAHs. Overall, the heatmap is consistent with an ecological model in which functional partitioning is driven by environmental gradients: metal/nutrient dynamics in Roodepoort favour varied, robust consortia, while severe PAH/acidity stress in Midrand may favour specialized aromatic degraders.

To further explore the relationship between microbial abundance and environmental stressors, we charted the mean relative abundance of the top 20 taxa versus their mean absolute correlation strength with essential physicochemical parameters (Figure 9). Kendall rank correlations between PAHs (AE = acenaphthene, IP = indeno[1,2,3-cd]pyrene, GHI = benzo[ghi]perylene, Anb = anthracene, BaA = benz[a]anthracene, Cnb = chrysene, Nnb = naphthalene), heavy metals (As = arsenic, Ni = nickel), and macronutrients (nitrogen) are integrated in this scatter plot. Points are arranged along the y-axis by mean absolute τ values (0.2–0.8) and the x-axis by average abundance (~0.01–0.6). Labels for each taxon indicate the two most strongly associated factors, such as strength and directionality (positive or negative).

A broad positive pattern can be seen in the plot: more abundant species (such as *Bacillus cereus* and *Pseudomonas citronellolis*) typically show larger overall environmental associations (mean $|\tau| > 0.5$), suggesting that site-specific pollutants may disproportionately shape dominant populations. Despite their rarity, a few of the low-abundance taxa (such as *Bacillus cereus* and *Williamsia muralis*) exhibit notably high correlation strengths ($|\tau|$ up to 1.0), suggesting potential specialized niche roles.

Notable environmental linkages and high abundance were both indicated by the dominant taxa's clustering in the upper-right quadrant of the plot. For instance, *Pseudomonas citronellolis* (mean abundance ~0.55) exhibited the highest mean absolute correlation (~0.75), with notable negative associations to As ($\tau = -1.0$) and Nnb ($\tau = -0.60$) but positive associations to AE ($\tau = 0.8$) and BaA ($\tau = 0.63$). This pattern is consistent with its enrichment in acidic, HMW-PAH-dominated soils in Midrand, where its MAG-encoded oxygenases may be active. Similarly, the Roodepoort prevalence of *Bacillus cereus* and *Bacillus thuringien-*

sis (abundance ~0.15–0.20 each) corresponds to the positive Pb associations previously mentioned, while their perfect negative association with AE and IP ($\tau = -1.0$) highlights a potential Firmicutes suppression under high PAH loads in Midrand. *Mycobacterium chimaera* (abundance ~0.10) showed a positive association with IP ($\tau = 0.8$) but a notable negative association with As ($\tau = -1.0$), highlighting actinobacterial tolerance for indenol-based PAHs but potential sensitivity to metalloids, which may explain its intermittent Midrand MAG detection.

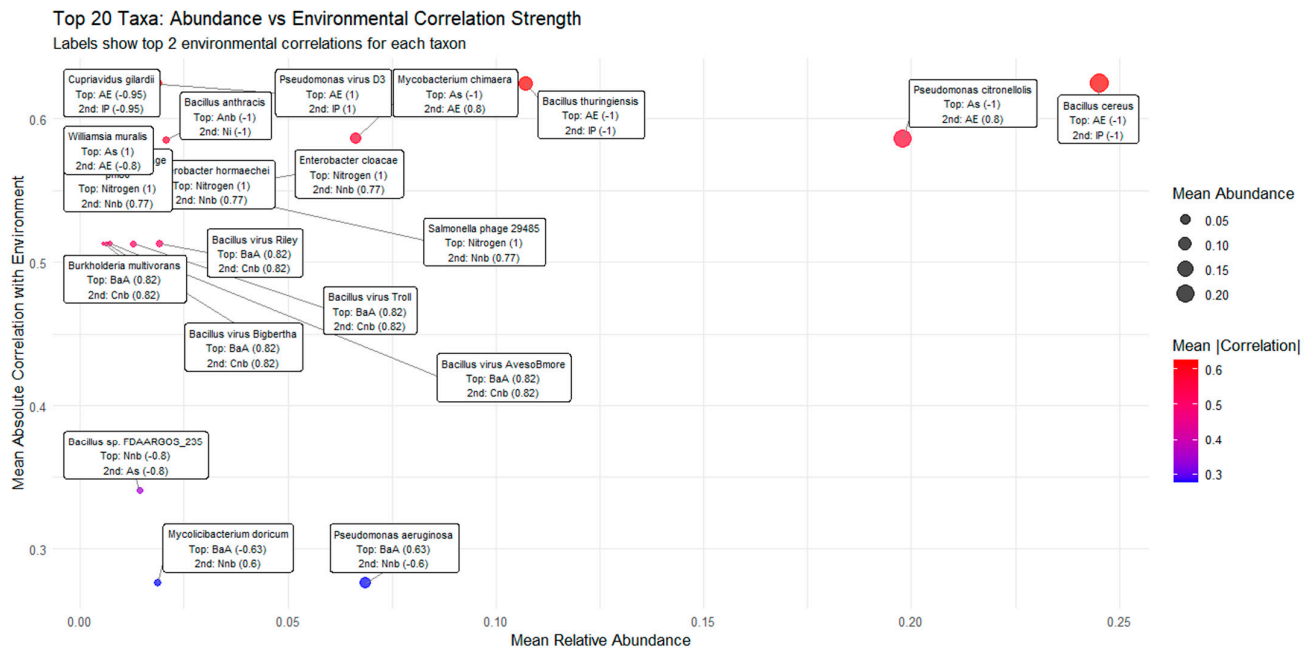


Figure 9. Top 20 taxa abundance vs. environmental correlation strength showing observed trends.

Lower-abundance taxa (less than 0.05) frequently inhabited the upper-left quadrant, exhibiting notable associations that suggest their roles in stress response, despite their scarcity. *Williamsia muralis*, an actinobacterium, exhibited a notable positive association with As ($\tau = 1.0$) and a potential strong negative association with AE ($\tau = -0.8$), identifying it as a potential metal bioaccumulator in the metal-enriched, lower-PAH soils of Roodepoort. *Bacillus anthracis* (closely related to *B. cereus*) exhibited negative associations with Anb and Ni ($\tau = -1.0$), indicating inhibition by anthracene-like PAHs and nickel, which may restrict its biosurfactant contributions. Phage entities such as *Enterobacteria* phage phi80 and *Salmonella* phage 29488 (abundance ~0.02–0.05) exhibited notable positive associations with nitrogen ($\tau = 1.0$) and Nnb ($\tau = 0.77$), suggesting a potential viral involvement in nutrient cycling and low-MW PAH dynamics, potentially influencing *Enterobacter* hosts in nitrogen-deficient Roodepoort samples.

Other notable patterns were *Bacillus* phages (*Bacillus* viruses *Bigbertha*, *Troll*, and *AvesoBmore*) exhibiting positive associations with BaA and Cnb ($\tau = 0.82$), suggesting a potential phage-mediated increase in PAH tolerance in *Bacillus* communities. *Cupriavidus gilardii* (abundance ~0.08) exhibited notable negative associations with AE and IP ($\tau = -0.95$), underscoring its position as a metal specialist (positive As/Pb associations) in initial PAH degradation. *Burkholderia multivorans* and *Mycolicibacterium doricum* demonstrated mixed patterns, showing positive BaA/Cnb ($\tau = 0.82$ – 0.63) and negative Nnb ($\tau = -0.60$), suggesting mid-ring PAH versatility. *Enterobacter cloacae* and *Enterobacter hormaechei* exhibited a positive correlation with nitrogen ($\tau = 1.0$) and Nnb ($\tau = 0.77$), highlighting their roles in nitrogen metabolism within Roodepoort consortia. *Pseudomonas* virus D3 exhibited perfect positive correlations with AE and IP ($\tau = 1.0$), potentially enhancing *Pseudomonas* PAH

degradation, but *Bacillus* sp. FDAARGOS_235 demonstrated negative associations with Nnb and As ($\tau = -0.80$), highlighting the potential sensitivity of the *Bacillus* sp.

Because uncommon taxa can have a disproportionate impact through specific adaptations, this visualization highlights an important ecological fact in these urban contaminated soils: abundance does not always correspond with environmental responsiveness. Positive metal and nutrient linkages for rarer actinobacteria and phages may indicate unrealized bioremediation potential in Roodepoort, while negative PAH correlations among abundant taxa such as *Bacillus* sp. and *Pseudomonas citronellolis* may reflect toxicity thresholds favouring *Pseudomonas* in Midrand. By combining these patterns with MAG functional profiles, a functional redundancy hypothesis is supported, in which dominant degraders may be complemented by low-abundance specialists (such as As-tolerant *W. muralis*).

3.4.1. Observed Negative Trend

There is a body of field and lab evidence that supports the observed negative trends between *Bacillus* species and elevated levels of metals and PAHs. The relative abundance of Firmicutes appears to be suppressed by excessive PAH pollution, according to several studies. For example, PAH contamination considerably decreased Firmicutes, Bacteroidota, and Desulfobacterota while increasing Acidobacteriota and Gemmatimonadota in the Yellow River Delta tidal flats [112]. Observations in mangrove wetlands, where Firmicutes were similarly repressed at high PAH concentrations, further support the observation that *Bacillus*-like Firmicutes may be inhibited in PAH-rich settings [112]. The enrichment of more resistant Proteobacteria contrasts with this inhibition. In a study by Yang et al. [113], *Pseudomonas* was positively associated with PAH content in steel-plant soils contaminated with PAHs, which is consistent with our finding that *Pseudomonas* may be more advantageous than *Bacillus* in highly contaminated layers.

At a more specific taxonomic level, research shows that certain degraders, like *Bacillus* and *Mycobacterium*, may be more common in soils with lower levels of low-molecular-weight (LMW) PAHs [114]. This suggests that these soils are more likely to be affected by severe or high-molecular-weight (HMW)-dominated pollution. Laboratory experiments demonstrate that metals such as Cd, Ni, and Zn impede the growth and hydrocarbon-degradation ability of both *Bacillus* and *Pseudomonas* at sub-minimum inhibitory concentrations, with inhibition correlating to metal dosage [115,116]. This evidence lends support to the interpretation that the notable negative trends seen for *B. cereus* and *Cupriavidus gilardii* with metals may indicate genuine physiological inhibition under mixed contaminant stress, rather than neutral relationships. The literature collectively indicates a consistent trend in contaminated ecosystems where elevated PAH and metal concentrations inhibit Firmicutes, including *Bacillus*, while favouring more resilient Proteobacteria, such as *Pseudomonas* [112–114,117].

3.4.2. Observed Positive Trends

In contrast, the positive relationships identified for specific taxa are supported by established adaptation mechanisms. Strains of the *Bacillus cereus* group, although potentially inhibited by PAHs, are extensively documented to demonstrate significant tolerance and biosorptive capacity for lead (Pb). Isolates from contaminated areas have shown the capacity to accumulate over 80% of lead through surface biosorption, corroborated by genetic evidence for heavy-metal resistance transporters and efflux systems [118,119]. Related species like *B. tequilensis* exhibit significant lead tolerance and immobilisation via exopolysaccharide-mediated biosorption and mineral precipitation [120], consistent with the noted associations between lead and *B. cereus* found.

The notable positive trend between *Enterobacter cloacae* and *Enterobacter hormaechei* and soil nitrogen is substantiated by their recognised function as plant-growth-promoting bac-

teria, proficient in nitrogen fixation, heterotrophic nitrification, and ammonia production, thereby augmenting soil nitrogen availability [121–123]. The positive associations between bacteriophage taxa and nitrogen, as well as PAHs, indicate a possible regulatory function in nutrient and pollutant cycles. Recent studies indicate that phages can substantially influence soil nitrogen fixation by lysing essential bacterial hosts such as *Enterobacter* [124] and may also transport metabolic genes that improve phenanthrene degradation by their hosts [125], thereby supporting the hypothesis that phage–host interactions may regulate bioremediation efficacy. The favourable correlation of *Cupriavidus gilardii* with arsenic and lead aligns with its designation as a prototypical metal-resistant bacterium, frequently possessing genes for metalloids oxidation and sequestration [126]. The association of the actinobacterium *Williamsia muralis* with arsenic corresponds with extensive research indicating that Actinobacteria are often predominant, metal-tolerant phyla in polluted soils, associated with metal immobilization [127]. The varied sensitivity of *Mycobacterium* to polycyclic aromatic hydrocarbons and metals is thoroughly characterised. Certain *Mycobacterium* species exhibit sensitivity to elevated contaminant levels, while others are identified as principal PAH-degrading taxa, with their abundance frequently rising along PAH contamination gradients [23,128]. This coincides with our discovery that various *Mycobacterium* taxa respond differentially to pollutants, suggesting niche specialization within this genus for more acidic, PAH-rich environments.

3.5. Functional Potential and Bioremediation Implications

The ability of the microbial communities in the oil-contaminated soils to degrade hydrocarbons was revealed in detail by genome-resolved functional annotation using DRAM on the medium- and high-quality metagenome-assembled genomes (MAGs) (Figure 10). For important catabolic gene categories involved in the breakdown of polycyclic aromatic hydrocarbons (PAH) and alkanes, such as transport systems, stress-response genes, oxygenases, oxidoreductases, monooxygenases, efflux pumps, dioxygenases, and cytochrome P450 enzymes, the heatmap displays the completeness percentage and normalized gene counts (ranging from 0 to over 250, with darker red denoting higher abundance). To highlight site-specific enrichment patterns that correspond with the physicochemical profiles (acidic, HMW-PAH-dominated Midrand vs. neutral-pH, metal-enriched Roodepoort), MAGs are labelled by sample origin, bin number, taxonomic order, and completeness.

Across the visualized MAGs, transport genes that are involved in aromatic compound uptake were consistently the most prevalent category, with up to 250 gene copies observed in high-completeness *Pseudomonas*-dominated MAGs from Midrand (Bin1_Mid_10_Pseudomonadales and Bin1_Mid_20_Pseudomonadales, both exceeding 94% completeness) and a few Roodepoort bins (Bin1_Rood_20_Bacillales at 65.6%, Bin2_Mid_10_Enterobacteriaceae at 99.69%, and Bin5_Rood_10_Bacteria at 65.52%). This highlights the essential function of substrate acquisition in hydrocarbon-degrading populations, facilitating efficient PAH solubilization and import under contaminant stress. Conversely, stress-response genes (such as chaperones and antioxidants) were consistently low (<50 copies in most MAGs, including *Pseudomonas* and *Bacillus* bins), indicating that, although these communities are adapted to persistent pollution, oxidative stress management may depend more on community-level interactions than on genome-encoded redundancy.

In Actinomycetales-affiliated MAGs from Midrand, Bin2_Mid_20_Actinomycetales at 98.34% and Bin3_Mid_10_Actinomycetales at 90.37%, oxygenases, oxidoreductases, and monooxygenases showed moderate enrichment. *Pseudomonas* and *Enterobacteriaceae* MAGs showed similar enrichment at <100 and <50 copies, respectively, for efflux pumps and dioxygenases, suggesting strong lower-pathway capabilities for funneling degraded aromatics into central metabolism. The MAGs derived from Roodepoort showed lower overall gene

counts, which is consistent with higher metal stress and milder PAH burdens. For instance, the Bacillales bins (Bin1_Rood_10 at 58.52% and Bin1_Rood_20 at 65.6%) had transport genes at 100–250 copies but minimal representation (<20–50) in oxidoreductases, monooxygenases, efflux, and oxygenases, suggesting a reliance on biosurfactant production rather than direct catabolism. Meanwhile, the Burkholderiales-affiliated Bin6_Rood_10 (34.26% complete) displayed intermediate levels (transport at 150, oxygenases/oxidoreductases less than 100, monooxygenases/efflux less than 50), highlighting *Cupriavidus*-like taxa as potential adaptable metal-PAH co-degraders.

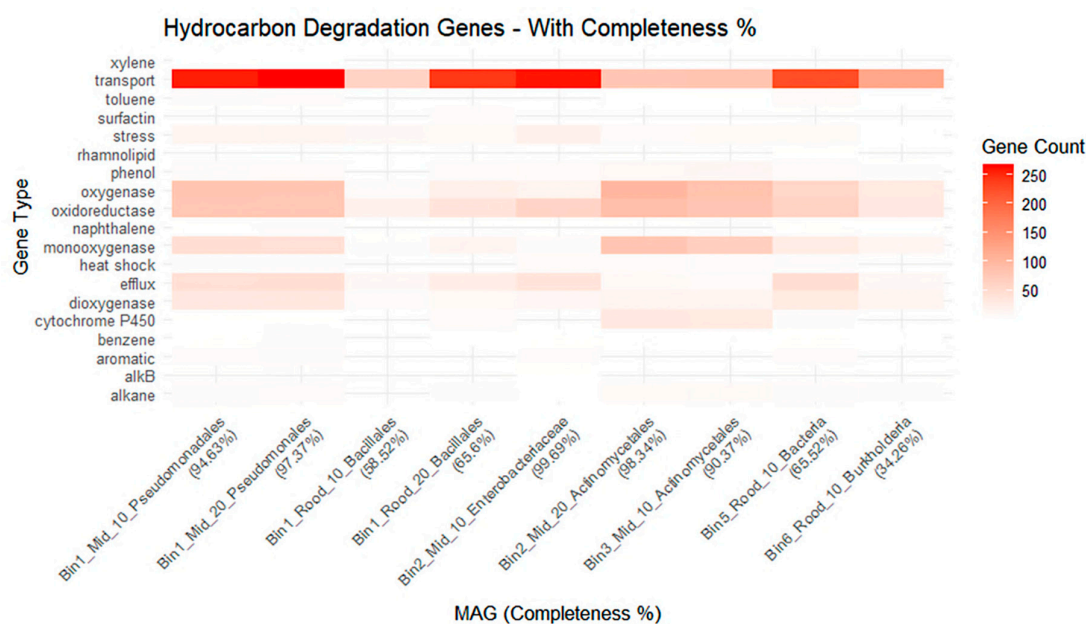


Figure 10. Genome-resolved completeness of hydrocarbon degradation genes in metagenome-assembled genomes (MAGs). Heatmap showing the normalized gene counts for key functional categories, including transporters, oxygenases, oxidoreductases, monooxygenases, dioxygenases, efflux pumps, aromatic catabolic enzymes, and biosurfactant genes. MAGs are labelled by sample site, bin number, taxonomic order, and completeness. Darker red indicates higher gene abundance. This visualization highlights site-specific enrichment patterns, with Midrand MAGs (Pseudomonadales, Actinomycetales) showing elevated upper-pathway PAH degradation genes and Roodepoort MAGs (Bacillales, Burkholderiales) showing high transport and metal-resistance genes.

Complementing this, the distribution of high-value petroleum degradation genes among the MAGs concentrated on categories that are essential for bioremediation: alkane degradation enzymes (aliphatic hydrocarbon breakdown), oxygenases (initial hydrocarbon activation), biosurfactants (improving bioavailability), and aromatic degradation enzymes (PAH catabolism) (Figure 11). The heatmap shows the normalized gene counts for these categories in a few high-quality MAGs. The counts range from about 20 to more than 200, with darker red denoting greater abundance.

Oxygenase genes were the most variably distributed category, reaching peak counts of >200 in several Midrand bins (Bin1_Mid_10_Pseudomonadales, Bin1_Mid_20_Pseudomonadales, Bin2_Mid_20_Actinomycetales, Bin3_Mid_10_Actinomycetales) and one Roodepoort bin (Bin5_Rood_10_Bacteria). The enrichment of oxygenases in Pseudomonadales and Actinomycetales MAGs in Midrand corresponds with the site's elevated HMW-PAH levels (e.g., benzo[a]pyrene at 8.5 µg/g), where multicomponent ring-hydroxylating oxygenases (RHOs) likely play a vital role in beginning degradation. Conversely, Roodepoort Bacillales bins (Bin1_Rood_10 and Bin1_Rood_20) demonstrated significantly reduced oxygenase counts (<50–100), aligning with the predominance of 4-ring PAHs (e.g., pyrene), which

may necessitate fewer specialized initiators. The Burkholderiales-associated Bin6_Rood_10 exhibited intermediate oxygenase levels (~100), indicating that *Cupriavidus*-like taxa play a small role in aromatic activation within metal-stressed environments.

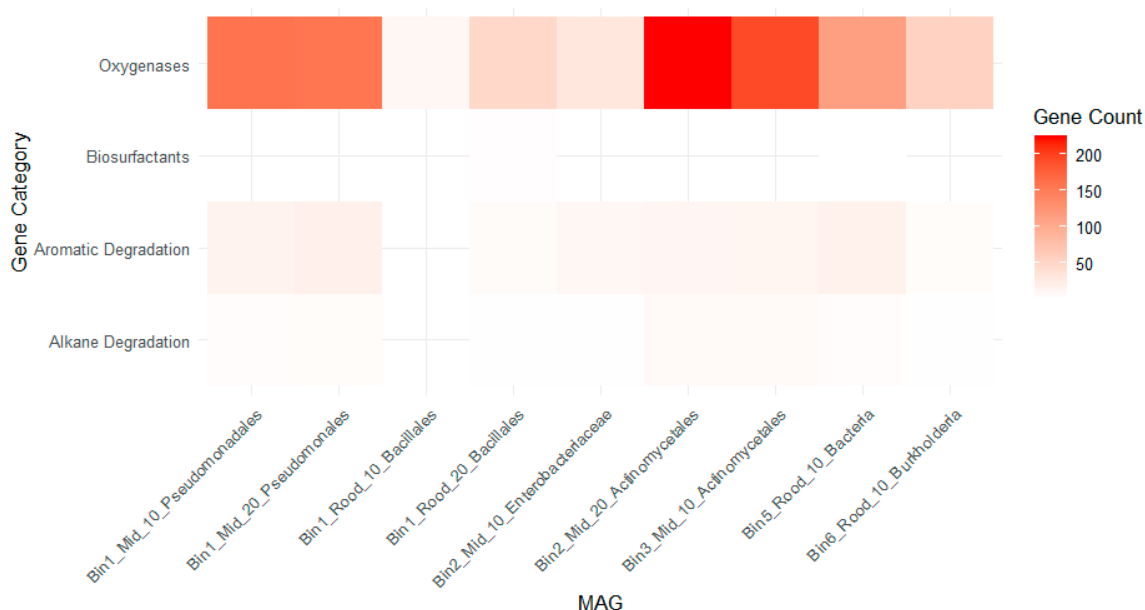


Figure 11. Distribution of high-value petroleum degradation genes across metagenome-assembled genomes (MAGs). Heatmap representing the abundance of genes involved in aliphatic hydrocarbon degradation, aromatic degradation, oxygenases, and biosurfactant production. MAGs are organized by completeness and taxonomic affiliation. This figure emphasizes functional partitioning, showing that high-completeness, high-abundance MAGs from Midrand carry extensive PAH-catabolic genes, whereas Roodepoort MAGs focus on biosurfactant synthesis and metal tolerance.

Except for faint signals in Roodepoort bins (e.g., Bin1_Rood_20_Bacillales), biosurfactant genes were sparsely represented across all MAGs, with minimal counts (<50, frequently near zero). This suggests that emulsification of hydrocarbons may be a niche function primarily carried out by *Bacillus* populations. In most bins, such as Pseudomonadales (Bin1_Mid_10 and Bin1_Mid_20), Actinomycetales (Bin2_Mid_20), and Enterobacteriaceae (Bin2_Mid_10), aromatic degradation genes (e.g., catechol dioxygenases, protocatechuate branches) were consistently low (<20–50). This suggests that while upper-pathway activation is potentially robust in Midrand, downstream funnelling to central metabolism depends on community synergy rather than single-genome completeness. Even smaller portions (<50, frequently insignificant) of alkane degradation genes were seen, especially in Pseudomonadales bins, suggesting a possible restriction in the digestion of aliphatic hydrocarbons across sites.

To determine or ascertain the microbial communities' capacity for bioremediation, an integrated evaluation of metagenome-assembled genome (MAG) quality, community abundance, and hydrocarbon-degradation potential was carried out (Figure 12). With bubble size representing relative abundance in the community (0.0–0.8) and colour indicating taxonomy, the bubble plot displays MAG completeness (%) on the x-axis (ranging from roughly 30% to 100%) and the number of degradation genes (such as oxygenases, oxidoreductases, monooxygenases, dioxygenases, efflux pumps, and aromatic catabolic enzymes) on the y-axis (ranging from roughly 100 to 600). This data suggests that high-completeness MAGs from predominant taxa typically contain a greater number of degradation genes, exhibiting site-specific clustering. Taxa linked with Midrand (e.g., Pseudomonadales, Actinomycetales) are positioned in the quadrant characterized by high completeness and

high gene count, whereas taxa from Roodepoort (e.g., Bacillales, Burkholderiales) exhibit varying completeness but frequently demonstrate high abundance.

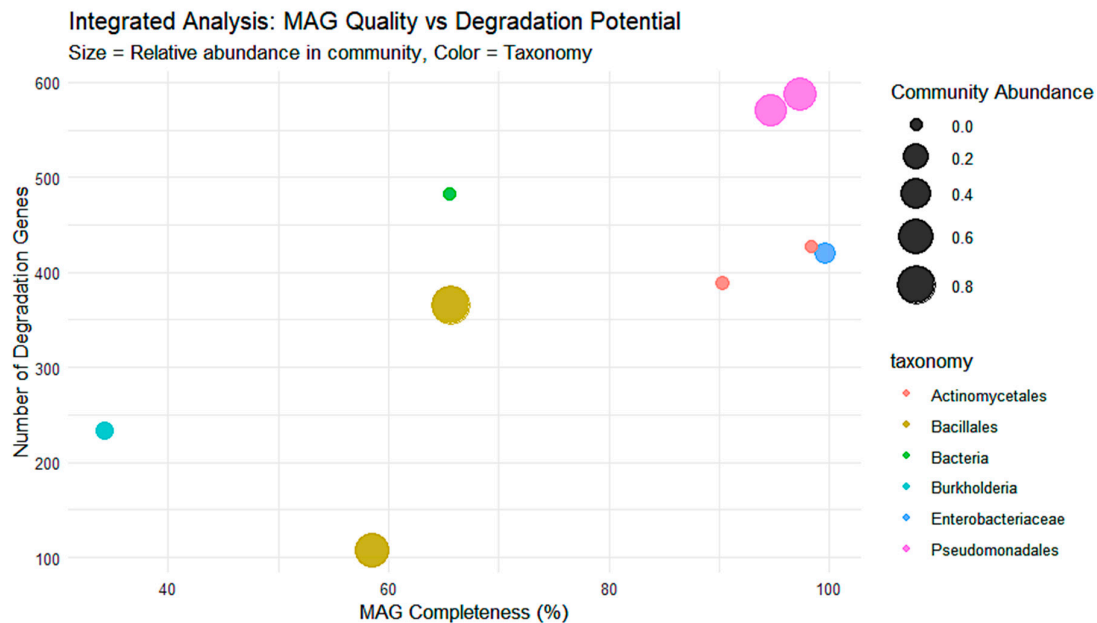


Figure 12. Integrated analysis of metagenome-assembled genome (MAG) quality, relative community abundance, and hydrocarbon-degradation potential.

Pseudomonadales MAGs, prevalent in Midrand samples, demonstrated outstanding quality and functional diversity, with completeness above 95% and almost 600 degradation genes, alongside significant relative abundance (0.6–0.8). This suggests that these genomes (presumably *Pseudomonas* sp.) are nearly complete and encode extensive systems for PAH degradation, encompassing ring-hydroxylating oxygenases and downstream pathways. Actinomycetales MAGs (*Mycobacterium*/*Mycolicibacterium*) from Midrand also demonstrated high gene counts (>350 and ~400, respectively) and completeness (first >90%, second ~100%), but at lower abundance (<0.2), suggesting potential specialized roles in HMW-PAH degradation despite their rarity. Enterobacteriaceae MAGs, such as *Enterobacter*, attained nearly 100% completeness with >400 degradation genes in the acidic, PAH-heavy soils of Midrand, despite their low abundance (0.0–0.2), potentially carrying out auxiliary functions.

In contrast, Bacillales MAGs (*Bacillus* spp.) from Roodepoort had moderate completeness (first ~50%, second >65%) but high degradation gene counts (>100 and >350, respectively) and high abundance (0.6–0.8 for the first, ~0.8 for the second). This indicates that they are dominant and resilient in environments with high metal levels and neutral pH. Burkholderiales MAGs (*Burkholderia*/*Cupriavidus*) exhibited reduced completeness (<40%) and gene counts of more than 200 at minimal abundance (0.0–0.2). In contrast, a general 'Bacteria' MAG demonstrated more than 60% completeness with approximately 500 genes, yet similarly low abundance (0.0–0.2). This indicates that these taxa may offer metal-tolerance support rather than serving as primary agents of hydrocarbon degradation. These patterns show that there are trade-offs in ecology: Midrand's high-quality Pseudomonadales and Actinomycetales MAGs prioritize gene-rich PAH catabolism, allowing for efficient bioremediation of resistant HMW-PAHs under stress. However, low-abundance specialists may limit scalability without enrichment. Roodepoort has a lot of Bacillales MAGs that, even though they are not very complete, have strong biosurfactant and efflux capacities (in addition to lower oxygenase counts). This makes them relevant for biostim-

ulation in areas that are co-contaminated. The relationship between completeness, gene count, and abundance supports functional partitioning.

DRAM analysis reveals site-specific functional adaptations in microbial communities. Midrand MAGs (Pseudomonadales, Actinomycetales) carry high counts of oxygenases, oxidoreductases, and transporters, consistent with HMW-PAH degradation. Roodepoort MAGs (Bacillales, Burkholderiales) show moderate catabolic genes but high transport and biosurfactant-related genes, reflecting adaptation to metal-enriched, lower-PAH soils. Overall, a few high-quality MAGs drive hydrocarbon catabolism, supported by other taxa performing auxiliary or stress-tolerance functions, highlighting functional partitioning relevant for bioremediation.

3.5.1. Genomic Insights into Hydrocarbon Degradation and Bioremediation Potential

According to several studies, oil contamination favours microbial consortia that have multi-step oxidative pathways and high-capacity uptake systems for hydrophobic substrates, which facilitates the effective mineralization of complex mixtures [129–131]. The genome sequencing of hydrocarbon degraders from contaminated soils, which showed extensive repertoires of dioxygenases, monooxygenases, and dehydrogenases supporting the degradation of benzoate, BTEX, naphthalene, and other aromatics, is an example of this pattern. *Pseudomonas* species often carry among the richest catabolic gene inventories [129]. Similarly, metagenomic investigation of a diesel-degrading consortium revealed a variety of ring-hydroxylating dioxygenases spread across several genera, with *Pseudomonas* predominating under alkane substrates, as well as redundant alkane monooxygenases [131]. Genes like *alkB*, alcohol dehydrogenase, and protocatechuate dioxygenases are consistently enriched in community-level predictions in oil-polluted soils, suggesting a greater ability for initial oxidation and aromatic ring cleavage [130,132].

The prevalence of aromatic transport systems in the *Pseudomonas*-dominated metagenome-assembled genomes (MAGs) from Midrand corresponds with transcriptome and metagenomic data that designate transporters and ABC systems as critical rate-limiting factors in polycyclic aromatic hydrocarbon (PAH) degradation [129,133]. These systems enhance the absorption of weakly soluble substrates, illustrating why PAH-degrading strains frequently upregulate transporter genes in response to hydrocarbon stress. Moreover, the augmentation of stress-response genes and efflux pumps in the MAGs aligns with research indicating that oil contamination promotes genetic modules for xenobiotic resistance, oxidative stress alleviation, and biofilm development, collectively facilitating microbial survival in toxic environments [17,134,135]. The divergent functional patterns of the acidic, HMW-PAH-dominated Midrand soils and the neutral, metal-enriched Roodepoort soils reflect findings that local physicochemical conditions influence both catabolic gene profiles and the taxonomic identities of degraders [17,136]. In various contaminated locations, areas exhibiting increased total petroleum hydrocarbon (TPH) and PAH concentrations or salinity demonstrate enhanced pathways for xenobiotic and aromatic degradation, frequently associated with a transition towards resilient degraders such as *Pseudomonas*, *Achromobacter*, and *Acinetobacter* [129,137].

The large copy counts of oxidative enzymes and transporters in Pseudomonadales MAGs indicate a significant potential for in situ biostimulation at Midrand from the standpoint of bioremediation. Similar to consortium-based or rhizoremediation techniques, functionally redundant catabolic networks generate significant TPH reductions; optimising nutrients, oxygen, and moisture could speed up PAH [131,137]. According to Roodepoort, measures that specifically address metal toxicity, like adjustments to lower metal bioavailability and prevent suppression of native degraders, are supported by the co-occurrence of hydrocarbon catabolism and metal-resistance genes. Overall, the genome-resolved profiles

show that both sites have native communities with significant intrinsic bioremediation capacity; to fully realise their potential, site-specific management is needed.

3.5.2. Functional Potential and Stress Adaptation

Although petroleum contamination primarily selects for oxygenases and ring-cleavage enzymes rather than large genomic expansions of classical oxidative-stress machinery, the relatively low copy numbers of generic stress-response genes (e.g., chaperones, antioxidants) in many MAGs are consistent with studies that show strong enrichment of hydrocarbon catabolic functions [138,139]. Strong enrichment of alkane/PAH degradation genes with only slight changes in generic stress-response markers is reported by metagenomic studies of oil-contaminated soils, indicating that oxidative stress is frequently buffered at the community or regulatory level [138,140]. Strong inducible antioxidant responses during PAH metabolism are highlighted by transcriptome investigations of certain degraders (e.g., *Rhodococcus*, *Debaryomyces*), suggesting regulatory plasticity over constitutive, redundant gene inventories [141]. The known function of actinomycetes (such as *Mycobacterium* and *Rhodococcus*) as experts in breaking down HMW aromatics and solvents, frequently carrying specialised mono-/dioxygenases, is consistent with the moderate enrichment of oxygenases and oxidoreductases in Midrand Actinomycetales MAGs [139,142]. The broad metabolic networks found in cultured degraders and metagenomes from petroleum-impacted sites, where catechol and protocatechuate routes predominate, are reflected in the similar distributions of peripheral and central aromatic pathway genes in *Pseudomonas* and *Enterobacteriaceae* MAGs [129,138].

The lower overall catabolic gene counts in Roodepoort MAGs, along with site chemistry (reduced PAHs, elevated metals), align with research indicating that metal stress limits hydrocarbon-degradation gene repertoires and alters communities towards taxa that prioritise metal resistance, biosurfactant synthesis, and resource acquisition [143,144]. From this perspective, Roodepoort Bacillales bins, characterised by numerous transporters and a small enrichment of monooxygenases, align with findings that indicate *Bacillus* typically plays a role in biosurfactant synthesis and emulsification rather than functioning as a major high-capacity degrader [145,146]. In contrast, the Burkholderiales-affiliated bin aligns with genomic studies indicating that environmental Burkholderiales may harbour vast aromatic ring-cleavage pathways and function as versatile metal-PAH co-degraders [147]. These patterns indicate specific bioremediation processes. In Midrand, interventions must focus on enhancing oxygen and nutrition availability to leverage the current deep catabolic potential, while in Roodepoort, a synergistic strategy involving metal immobilisation and the activation of essential degraders and biosurfactant producers is recommended.

3.5.3. Distribution of High-Value Petroleum Degradation Genes

Numerous studies have identified ring-hydroxylating dioxygenases (RHDs) as the crucial, frequently rate-limiting step in PAH degradation [148,149]. This is consistent with the heterogeneous enrichment of oxygenase genes across MAGs, with the highest counts in Midrand Pseudomonadales and Actinomycetales. According to metagenomic research, petroleum contamination promotes ring opening and the build-up of downstream intermediates via upregulating the monooxygenase and dioxygenase systems [149]. Additionally, PAH content induces enrichment of RHD-harboring taxa, with a larger loading preferring increased dioxygenase gene abundance, according to gene-targeted surveys [150]. Whole-genome investigations at the single-strain level show extensive oxygenase repertoires in genera like *Pseudomonas* and *Rhodococcus*, which facilitate the breakdown of several PAHs through catechol/protocatechuate pathways [129].

The co-occurrence of alkane and aromatic catabolic genes across MAGs indicates the extensive dual degradation capability observed in oil-associated environments [131,150]. This is supported by the presence of biosurfactant genes, which are closely associated with the capacity to degrade alkanes and PAHs by enhancing substrate bioavailability [151]. The incorporation of biosurfactant, transporter, and oxygenase genes within essential MAGs signifies a functionally integrated mechanism for addressing low solubility and initiating oxidation. The contrast between Roodepoort and Midrand exemplifies a broader ecological pattern whereby the ability to degrade hydrocarbons is functionally segregated. Only a subset of genomes possesses extensive catabolic repertoires, whereas others contribute specialised functions, such as emulsification or stress tolerance [152,153]. In Roodepoort, the low to intermediate oxygenase counts and minimal biosurfactant signals indicate a community where a limited number of specialists coexist alongside supporting organisations. In Midrand, the elevated abundance of upper pathway oxygenases, coupled with fewer counts of central funnel genes per MAG, supports a consortium model in which complete mineralization necessitates metabolic handoffs among different taxa [129,154].

3.5.4. Integration of Genomic Features (MAG Completeness, Gene Load, and Abundance)

The joint consideration of MAG completeness, hydrocarbon-degradation gene load, and abundance aligns with genome-resolved studies where high-quality, dominant MAGs typically carry the densest catabolic repertoires [155,156]. The clustering of Midrand *Pseudomonadales* MAGs in the high-completeness, high-gene-count, and high-abundance quadrant echoes work identifying *Pseudomonas* as a primary degrader with exceptionally broad aromatic pathways [129,157]. In contrast, the Midrand *Actinomycetales* (e.g., *Mycobacterium*) MAGs, with high completeness and substantial gene counts but low abundance, align with their recognized role as low-abundance specialists for HMW-PAH degradation [158,159]. The nearly complete *Enterobacteriaceae* MAGs with lower abundance are consistent with the findings where *Enterobacter* can provide auxiliary functions like nitrogen cycling and co-metabolism [155]. Roodepoort MAGs, displaying more variable completeness, higher abundance, but fewer total degradation genes, fit patterns where abundant taxa contribute supportive niche functions rather than serving as the main catabolic bacteria [152,155]. Overall, this structure, in which bioremediation capability is concentrated among a limited number of adaptable, high-quality taxa and supported by a broader guild of specialists, reflects previous MAG-resolved hydrocarbon systems and highlights the site-specific arrangement of degradation potential. This indicates that Midrand communities are better suited for PAH degradation, but Roodepoort's ability may depend more on interspecies collaboration under metal stress.

4. Limitations and Future Perspectives

The pooling of soil samples for metagenomic sequencing yielded a restricted effective sample size ($n = 4$), hence limiting the statistical power of the results. Thus, the associations and correlations outlined are exploratory and hypothesis-generating, reflecting observable co-occurrence patterns rather than statistically significant causal connections. Moreover, whereas metagenomics uncovers genetic potential, it does not directly assess *in situ* metabolic activity; the true bioremediation capability of these communities still requires experimental validation. The genome-resolved metagenomic analysis offers a comprehensive overview of the genetic potential for hydrocarbon breakdown and metal resistance in polluted soils; nonetheless, it is crucial to recognise that DNA-based analyses indicate genetic ability rather than metabolic activity. The existence of catabolic genes (e.g., oxygenases, dioxygenases, efflux pumps) does not ensure their expression, especially in environments contaminated with heavy metals, which may impede enzymatic activity

by toxicity, protein denaturation, or transcriptional repression. The elevated quantities of zinc and lead in Midrand, along with chromium and copper in Roodepoort, may inhibit the expression of PAH-degradation pathways, even in genetically predisposed taxa like *Pseudomonas* and *Bacillus*. Consequently, the shift from genetic potential to tangible bioremediation activity is merely an interpretation derived from genomic enrichment and association patterns. To validate the functional engagement of the discovered genetic machinery under in situ settings, future studies should carry out metatranscriptomics and/or metaproteomics to directly monitor and profile gene expression and protein synthesis from soil samples. These methods would clarify whether the suggested site-specific bioremediation strategies, such as pH modification in Midrand or biostimulation in Roodepoort, successfully initiate the degradation pathways encoded by the indigenous microbiome. Integrating multi-omics data with controlled microcosm experiments will connect potential and activity, facilitating more dependable and predictive bioremediation frameworks.

5. Conclusions

This exploratory study analyzed the physicochemical parameters and metagenomic potential of oil-contaminated soils from two different metropolitan locations in Gauteng, South Africa. This research shows that oil-contaminated soils in Midrand and Roodepoort host unique microbial communities shaped by differing physicochemical environments. Midrand's acidic, polycyclic aromatic hydrocarbon-rich, and metal-contaminated environment supported a consortium dominated by *Pseudomonas* species, possessing a wide array of catabolic genes for the degradation of high-molecular-weight hydrocarbons. In contrast, Roodepoort's predominantly neutral, metal-stressed soils favour a *Bacillus*-rich community with increased metal resistance and biosurfactant production capabilities. Analysis of observed trends between contaminant type, concentration, and soil pH serves as a key determinant of microbial community composition and functional capacity. The recovery of medium- to high-quality MAGs yielded genome-resolved insights into the genetic mechanisms underlying hydrocarbon degradation and stress response adaptation. The existence of specialized metabolic pathways, including those for PAH ring cleavage, heavy metal efflux, and biosurfactant production, underscores the inherent bioremediation capacities of native microbial communities. Nevertheless, the pronounced disparities among sites underscore that no single remediation strategy will be universally applicable.

These findings illustrate the importance of implementing site-specific bioremediation strategies. In Midrand, efforts should be directed towards improving the activity of indigenous *Pseudomonas* populations through pH regulation and nutrient supplementation to expedite the degradation of PAHs. In Roodepoort, the biostimulation of *Bacillus* spp., in conjunction with metal immobilization methods, may demonstrate greater efficacy. This research not only enhances our understanding of microbial ecology in co-contaminated urban soils but also offers a practical framework for utilizing native microbiomes in sustainable environmental restoration.

Supplementary Materials: The following supporting information can be downloaded at: <https://www.mdpi.com/article/10.3390/environments13030125/s1>. Table S1: MetaQUAST summary statistics of MEGAHIT assemblies (contigs ≥ 1 kb); Table S2: CheckM output; Figure S1: CheckM plot example for Mid_20 cm; Figure S2: Principal Coordinate Analysis (PCoA) comparing Beta Diversity across soil samples from Midrand and Roodepoort, illustrating differences in microbial community composition at various taxonomic levels: Phylum, genera, and species diversity.

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